

A nodal Gaussian obstruction to the adaptive fixed-point equation

Candidate exact counterexample and partial classification for arXiv:1805.00703

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Status. Candidate exact counterexample, likely valid. The source’s implicit adaptation equation is not well posed for every admissible Sobolev function. For $f(x) = xe^{-ax^2/2}$ it has no positive solution at $x = 0$ whenever $\lambda^2 \geq 2/3$, including $\lambda = 1$ in the source’s empirically favorable range. More generally, every Gaussian monomial has an exact sharp nodal threshold. This is a substantial partial answer, not a classification of all functions in the source’s open problem.

1 The source problem

Klebanov proposes an implicit, matrix-valued adaptation equation for $f \in W^{2,2}(\mathbb{R}^d, \mathbb{R}) \setminus \{0\}$ and $0 < \lambda < \sqrt{2}$. In one dimension, write $s(x) = \mu_f(x)^2 > 0$. The equation at a fixed spatial point x is

$$s = \frac{\lambda^2 s}{2} + \frac{1}{2} R_f(s; x), \quad (1)$$

where

$$R_f(s; x) = \frac{(f'^2 - f f'') * G_{(\lambda\sqrt{s})^{-2}}^2(x)}{f^2 * G_{(\lambda\sqrt{s})^{-2}}^2(x)}. \quad (2)$$

The corresponding iteration replaces s on the right by s_n and defines s_{n+1} . Remark 23 assumes that the equation has a unique positive solution and that the iteration converges to it from every positive initial value, then states that the class of functions for which this holds is open. The exact source passage is reproduced in Figure 1.

For the normalized one-dimensional Gaussian with covariance C , its square is proportional to $e^{-y^2/C}$. Therefore, at $x = 0$, the squared window in (2) contributes precisely the factor

$$\exp(-\lambda^2 s y^2); \quad (3)$$

its normalization cancels from the quotient.

2 Proof intuition

The source computes the fixed point for a nonvanishing Gaussian. A zero of the signal changes the calculation decisively because the denominator of (2) acquires extra powers of the integration variable. Gaussian moment ratios then grow linearly with s , feeding an additional multiple of s_n back into the iteration.

Remark 23. Here and in the following we will assume that the function f is such that (11) has a unique symmetric and positive definite solution μ_f and that the corresponding fixed point iteration

$$\mu_f^{(n+1)}(x) = \sqrt{\frac{(\lambda\mu_f^{(n)})^2(x)}{2} + \frac{(\nabla f \nabla f^\top - f D^2 f) * G_{(\lambda\mu_f^{(n)})^{-2}(x)}^2}{2 f^2 * G_{(\lambda\mu_f^{(n)})^{-2}(x)}^2}}(x)} \quad (12)$$

converges to that solution for every symmetric and positive definite choice $\mu_f^{(0)}$. The class of functions f for which this is the case remains an open problem.

The crucial advantage of the choice (11) is that it fulfills the Adaptation Axioms 14:

Theorem 24. Let $\mu_f^{(d)}$ be the solution of the implicit equation (11). Then it fulfills the Adaptation

Figure 1: Remark 23 and iteration (12), arXiv:1805.00703v1, PDF page 13.

For the first nodal Gaussian $f(x) = xe^{-ax^2/2}$, the nonlinear-looking scheme becomes the affine recurrence

$$s_{n+1} = \frac{3a}{2} + \frac{3\lambda^2}{2}s_n.$$

Its slope crosses one at $\lambda^2 = 2/3$. Below this value there is one positive fixed point and geometric convergence at the node. At the threshold the iterates drift linearly, and above it they grow exponentially; the only algebraic fixed point is then nonpositive. Thus a zero of a perfectly regular Schwartz function destroys even existence.

The same calculation works for $x^m e^{-ax^2/2}$. The consecutive even Gaussian moments give the exact threshold $(4m - 2)/(4m - 1)$. A further calculation away from the node shows that the first threshold is sharp for global pointwise solvability of the implicit equation, not merely an artifact of evaluating at one point.

3 Exact Gaussian-monomial classification at the node

Theorem 1. Let $a > 0$, $m \in \mathbb{N}$, and

$$f_m(y) = y^m e^{-ay^2/2}.$$

For the source iteration at $x = 0$, put $s_n = \mu_n(0)^2$. Then

$$s_{n+1} = A_m + r_m s_n, \quad A_m = \frac{a(4m - 1)}{2(2m - 1)}, \quad r_m = \frac{\lambda^2(4m - 1)}{2(2m - 1)}. \quad (4)$$

Consequently:

(i) if $\lambda^2 < (4m - 2)/(4m - 1)$, there is one positive fixed point,

$$s_* = \frac{a(4m - 1)}{2(2m - 1) - \lambda^2(4m - 1)},$$

and every $s_0 > 0$ converges to it with $s_n - s_* = r_m^n(s_0 - s_*)$;

(ii) at equality, no fixed point exists and $s_n = s_0 + nA_m$;

(iii) above the threshold, no positive fixed point exists and every positive orbit diverges exponentially.

For $m = 0$, the nonvanishing Gaussian instead gives $s_{n+1} = a/2 + (\lambda^2/2)s_n$, which converges throughout the source range $0 < \lambda < \sqrt{2}$.

Proof. For $m \geq 1$, direct differentiation (or differentiating $\log |f_m|$ away from zero and extending the polynomial identity) gives

$$f_m'^2 - f_m f_m'' = (ay^{2m} + my^{2m-2}) e^{-ay^2}. \quad (5)$$

At $x = 0$, combine this with (3) and put $c = a + \lambda^2 s$. If

$$I_j(c) = \int_{\mathbb{R}} y^{2j} e^{-cy^2} dy,$$

the standard integration-by-parts recurrence is

$$\frac{I_{m-1}(c)}{I_m(c)} = \frac{2c}{2m-1}.$$

The quotient in (2) is therefore

$$\begin{aligned} R_{f_m}(s; 0) &= a + m \frac{I_{m-1}(c)}{I_m(c)} \\ &= \frac{a(4m-1)}{2m-1} + \frac{2m\lambda^2}{2m-1} s. \end{aligned}$$

Substitution into (1) gives exactly (4). An affine recurrence with positive intercept has a positive fixed point and convergence from every positive start exactly when $r_m < 1$. The three cases and the displayed formulas follow immediately.

When $m = 0$, identity (5) reduces to $f_0'^2 - f_0 f_0'' = a f_0^2$, so $R_{f_0} = a$ independently of the window and the last assertion follows. $\square$

Corollary 2. *For every λ with*

$$\sqrt{2/3} \leq \lambda < \sqrt{2},$$

the admissible Schwartz function $f(x) = x e^{-ax^2/2}$ has no positive solution of the source equation at $x = 0$. Hence it has no positive adaptation function solving the equation on $\mathbb{R}$, and its fixed-point iteration cannot have the property required in Remark 23. In particular, this failure occurs at $\lambda = 1$, inside the source's reported favorable interval $[0.6, 1.2]$.

Proof. Take $m = 1$ in Theorem 1; its threshold is $2/3$. Failure at a single spatial point rules out a positive solution function on all of $\mathbb{R}$. The source's empirical interval is stated in its conclusion. $\square$

4 Sharp global solvability for the first nodal Gaussian

Proposition 3. *Let $f(y) = y e^{-ay^2/2}$. The implicit equation (1) has a unique positive solution $s = s(x)$ at every $x \in \mathbb{R}$ if and only if $\lambda^2 < 2/3$. In that regime $s(x)$, and hence $\mu_f(x) = \sqrt{s(x)}$, is continuous.*

Proof. Write $L = \lambda^2$, $b = Ls$, and $c = a + b$. Completing the square in the Gaussian integrals shows that, under their normalized weight, the integration variable has mean bx/c and variance $1/(2c)$. Since

$$f^2 = y^2 e^{-ay^2}, \quad f'^2 - ff'' = (1 + ay^2)e^{-ay^2},$$

we obtain the exact formula

$$R_f(s; x) = a + \frac{2c^2}{c + 2b^2x^2}. \quad (6)$$

For $L < 2/3$, define $F_x(s) = R_f(s; x) - (2 - L)s$. We have $F_x(0) = 3a > 0$. As $s \rightarrow \infty$, $F_x(s) \rightarrow -\infty$: for $x \neq 0$, the first term in (6) tends to $a + x^{-2}$, while for $x = 0$ one has $F_0(s) = 3a - (2 - 3L)s$.

It remains to prove uniqueness. If $h(b) = 2(a + b)^2/(a + b + 2b^2x^2)$, direct differentiation gives

$$h'(b) = \frac{2(a + b)(a + b - 4abx^2)}{(a + b + 2b^2x^2)^2} \leq 2.$$

Thus $\partial_s R_f \leq 2L$ and

$$\partial_s F_x \leq 3L - 2 < 0.$$

The intermediate value theorem and strict decrease give exactly one positive root. The displayed strict derivative bound and the smooth formula imply continuous dependence of the root on x . The converse follows from Corollary 2. $\square$

Remark 4. Proposition 3 classifies existence and uniqueness of the implicit equation for this test function. It does not assert convergence of the raw iteration at every nonzero x . The derivative of that map need not be a uniform contraction over all x and s , so a separate dynamical argument would be required. At the decisive node $x = 0$, Theorem 1 gives the complete iteration dynamics.

5 Verification, novelty, and review

The core calculation is exact: it uses only differentiation and the recurrence for even Gaussian moments. The companion script `code/verify_gaussian_monomial.py` symbolically verifies the $m = 1$ identity and all threshold formulas. No numerical approximation enters the proof.

A bounded search of the run's lightweight indexes and primary arXiv sources found the source paper and its companion on variable kernel density estimation, but no later treatment of this fixed-point obstruction. The source proves the constant fixed point for a nonvanishing Gaussian, but does not analyze the iteration or nodal Gaussians. Novelty confidence is moderate; exhaustive priority is not claimed.

This packet gives a full counterexample to universal well-posedness and an exact infinite-family threshold, but not the complete class requested by the open problem. A human reviewer should check the translation from the source's Gaussian covariance notation to (3), the Gaussian-moment quotient, and the derivative bound in Proposition 3.

References

- [1] I. Klebanov, *Adaptive Convolutions*, arXiv:1805.00703v1 (2018), especially equations (11)–(12), Remark 23 on PDF p. 13, and the parameter discussion on PDF p. 16.